NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

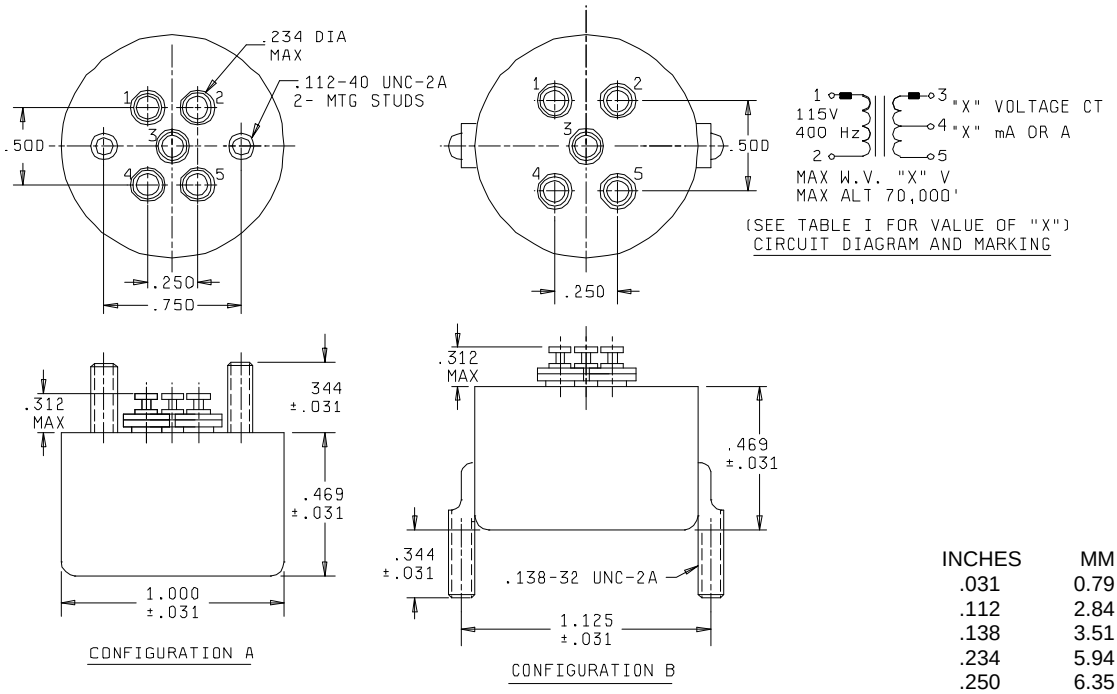
MIL-PRF-27/48C
7 February 1984
SUPERSEDING
MIL-T-27/48B
12 May 1978

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 1.5 VOLT AMPERES, 400 HZ, TF4S03YY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the transformers described herein shall consist of this specification and the latest issue of MIL-T-27.



INCHES	MM
.031	0.79
.112	2.84
.138	3.51
.234	5.94
.250	6.35
.312	7.92
.344	8.74
.469	11.91
.500	12.70
.750	19.05
1.000	25.40
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

© denotes changes

FSC 5950

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REQUIREMENTS (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings See table I.

Primary voltage (1-2) 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction

Dimensions and configuration See figure 1

Weight 1 ounce maximum.

Duty cycle continuous

Case Metal enclosed

Material steel

Altitude 000 feet, maximum.

Operating temperature range -55°C to +130°C.

Terminals and lead

Terminal height 0.312 inch maximum

Terminal strength Method 211 of MIL-STD-202, test condition A, 5 pounds.

Dielectric withstanding voltage

At sea level 2.5 times the maximum working voltage specified in table I.

At reduced barometric pressure .25 times the maximum working voltage specified in table I.

Electrical characteristics

No load With 115 volts, 400 Hz across (1-2) current in (1-2) shall not exceed 1.5 ma rms

Rated load With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I

Efficiency 70 percent minimum at rated load.

Polarity Additive with terminals 2 and 3 connected.

Regulation $\frac{\text{voltage (no load)} - \text{voltage (rated load)}}{\text{voltage (rated load)}} \times 100$

shall not exceed 23 percent

Phase shift 0 electrical degree between primary and secondary voltage at rated load maximum

Temperature rise 40°C, maximum with 115 volts, 400 Hz across (1-2) at an ambient temperature of 115°C

Vibration MIL-STD-202, method 204, test condition D

Marking location See figure

© QUALITY ASSURANCE PROVISIONS

Extent of qualification

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Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-T-27/48 through MIL-T-27/66 inclusive, all parts.

Qualification testing and approval to M27/48-44 shall be sufficient to grant qualification approval to M27/48-01 through M27/48-44.

Part number example

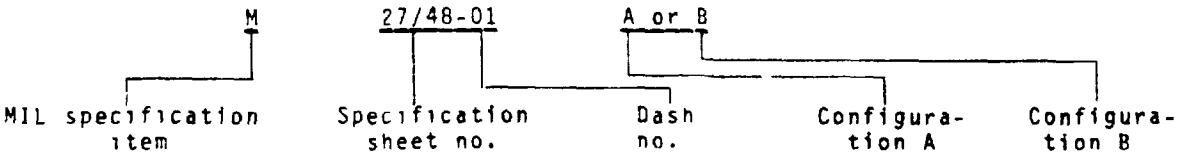


TABLE I. Electrical ratings.

Dash no.	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	6	0.250	325
02 "	6.3	0.238	"
03 "	7	0.214	"
04 "	7.5	0.200	"
05 "	8	0.188	"
06 "	9	0.167	"
07 "	10	0.150	"
08 "	11	0.136	"
09 "	12	0.125	"
10 "	12.6	0.119	"
11 "	13	0.115	"
12 "	15	0.100	"
13 "	16	0.094	"
14 "	18	0.083	"
15 "	20	0.075	"
16 "	22	0.068	"
17 "	24	0.063	"
18 "	26	0.058	"
19 "	28	0.053	"
20 "	30	0.050	"
21 "	33	0.045	"
22 "	36	0.042	"
23 "	39	0.038	"
24 "	43	0.035	"
25 "	47	0.032	"
26 "	51	0.029	"
27 "	56	0.027	"
28 "	62	0.024	"
29 "	68	0.022	"
30 "	75	0.020	"
31 "	82	0.018	"
32 "	91	0.016	"
33 "	100	0.015	"
34 "	110	0.014	"
35 "	115	0.013	"
36 "	120	0.0125	"
37 "	130	0.0115	"
38 "	150	0.010	"
39 "	180	0.0085	"
40 "	200	0.0075	440
41 "	220	0.007	440
42 "	240	0.006	440
43 "	270	0.0055	580
44 "	300	0.005	580

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Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities.
Army - ME
Navy - MC, AS
Air Force - 19

Agent.
DLA - ES

Preparing activity:
Army - ER

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